

Which Treatment Should We Choose?

A Bayesian comparison of three treatments

MA 232 Bayesian Statistical Methods · Group Project 2

Aditya Singh (250003007) Hiranish Anand (250041018) Parth Pawar (250041030)
Tirthanker Singh (250003081) Gulam Abbas (250041016)

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Summary. With independent uniform priors, the posterior success probabilities are $\theta_A \sim \text{Beta}(63, 39)$, $\theta_B \sim \text{Beta}(71, 31)$ and $\theta_C \sim \text{Beta}(67, 35)$. Treatment B has the highest posterior mean (0.6961). The posterior probability that B is better than A is 0.882, and that B is the best of the three is 0.681; both are computed exactly, not simulated. Choosing B has the smallest expected loss, 1.28 percentage points of success rate, and B is chosen under every prior we tried. We recommend **treatment B**, while noting that the evidence that B beats C is weak (probability 0.726).

1 Problem and data

The project sheet gives simulated clinical-trial data for three treatments (Table 1). It asks for: (1) the estimated success probabilities, (2) the differences between treatments, (3) the uncertainty in the estimates, (4) the Bayesian posterior distributions, (5) the probability that B is better than A, (6) the probability that B is the best treatment, and (7) a final recommendation. Sections 3–6 answer these in order. Section 7 checks the Module 5 approximations against our exact answers.

Table 1: Trial data (project sheet, Project 2).

Treatment	Patients n	Successes s	Failures $n - s$	Sample proportion s/n
A	100	62	38	0.62
B	100	70	30	0.70
C	100	66	34	0.66

Assumptions. Within a treatment, patients respond independently with a common success probability θ_k . The three groups contain different patients and are analysed independently. Each outcome is a simple success or failure. Section 8 discusses these assumptions.

2 Model

Likelihood and sufficiency. For one treatment, let $x_1, \dots, x_n \in \{0, 1\}$ be the outcomes, with $s = \sum x_i$. Each $x_i \sim \text{Bernoulli}(\theta)$, so

$$f(\mathbf{x} \mid \theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \underbrace{\theta^s (1 - \theta)^{n-s}}_{g_\theta(s)} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

By the Neyman–Fisher factorisation, s is sufficient for θ [Module 3, §2, p. 4]. Only the counts in Table 1 matter, not the order in which patients were treated. The likelihood is maximised at $\hat{\theta}_{\text{MLE}} = s/n$ [Module 3, p. 14].

Prior and posterior. With a $\text{Beta}(\alpha, \beta)$ prior, $\pi(\theta \mid \mathbf{x}) \propto \theta^{s+\alpha-1}(1-\theta)^{n-s+\beta-1}$, so

$$\theta \mid \mathbf{x} \sim \text{Beta}(s + \alpha, n - s + \beta) \quad [\text{Module 4, Example 4.6, pp. 4–5}].$$

Our main prior is the uniform $\text{Beta}(1, 1)$. It expresses no preference between success rates and treats the three treatments identically. It gives $\theta_A \mid \text{data} \sim \text{Beta}(63, 39)$, $\theta_B \mid \text{data} \sim \text{Beta}(71, 31)$ and $\theta_C \mid \text{data} \sim \text{Beta}(67, 35)$. Because the groups and priors are independent, the joint posterior is the product of the three, which makes the joint probabilities in Sections 4–5 computable. Section 6 repeats the key results under three other priors.

Bayes estimators. The point estimate depends on the loss function [Module 4, Theorems 4.1–4.3, pp. 7–8]. For a $\text{Beta}(a, b)$ posterior:

- squared-error loss gives the posterior mean $a/(a+b)$;
- absolute-error loss gives the posterior median, which has no closed form and is computed numerically;
- zero-one loss gives the posterior mode $(a-1)/(a+b-2)$.

3 Estimates, uncertainty and posterior distributions (questions 1, 3, 4)

Table 2: Point estimates and 95% intervals. CrI: equal-tailed credible interval from the Beta posterior. CI: classical Wald interval $\hat{\theta} \pm 1.96\sqrt{\hat{\theta}(1-\hat{\theta})/n}$.

	MLE	Post. mean	Post. median	Post. mode	Post. sd	95% CrI	95% Wald CI
A	0.62	0.6176	0.6184	0.6200	0.0479	(0.522, 0.709)	(0.525, 0.715)
B	0.70	0.6961	0.6974	0.7000	0.0453	(0.604, 0.781)	(0.610, 0.790)
C	0.66	0.6569	0.6579	0.6600	0.0468	(0.562, 0.745)	(0.567, 0.753)

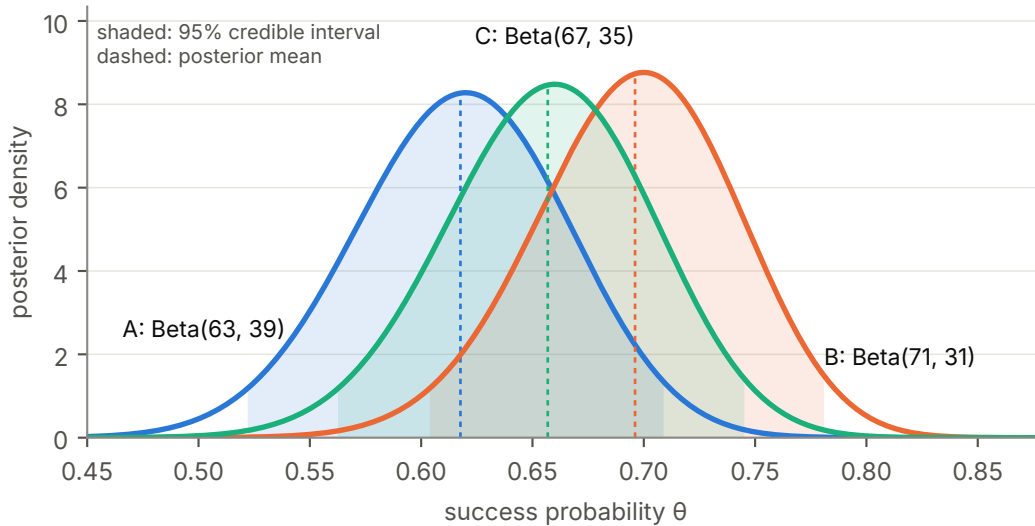


Figure 1: Posterior densities of the three success probabilities under uniform priors. The curves overlap substantially: B is ahead, but none of the three is clearly separated from the others.

Reading the estimates. The posterior mean of B is $\frac{71}{102} = 0.6961$. As in every conjugate model in the course, it is a weighted average of the MLE and the prior mean: $\frac{s+1}{n+2} = \frac{n}{n+2} \cdot \frac{s}{n} + \frac{2}{n+2} \cdot \frac{1}{2}$. With $n = 100$, the data weight is 0.980, so the prior moves each estimate only slightly towards $\frac{1}{2}$. Under the uniform prior the posterior mode equals the MLE. Each posterior has $\alpha > \beta$, so

it is left-skewed, and mean < median < mode, as Table 2 shows [the same ordering as Module 4, Example 4.9].

Uncertainty. Each posterior standard deviation is about 0.045–0.048, so each success rate is known to within roughly ± 9 percentage points (95%). The credible and Wald intervals almost coincide numerically, but they mean different things. The credible interval is a probability statement about θ given these data: $P(0.604 < \theta_B < 0.781 \mid \text{data}) = 0.95$. The Wald interval describes a procedure that covers the true θ in 95% of repeated trials.

Predictions for future patients. The probability that the next patient on a treatment succeeds is that treatment's posterior mean. For 100 new patients on B, the number of successes follows a beta-binomial distribution with mean 69.6 and standard deviation 6.4. Its 95% predictive interval is 56–82 successes; on the integer scale this actually covers 0.965. Predictive intervals are wider than credible intervals, because they add patient-to-patient variation to the uncertainty about θ_B .

4 Differences between treatments, and $P(\theta_B > \theta_A)$ (questions 2 and 5)

For a pair of treatments, the posterior of the difference, such as $\delta = \theta_B - \theta_A$, is the distribution of a difference of independent Beta variables. Its mean is the difference of the posterior means, its variance is the sum of the variances, and its quantiles come from

$$P(\theta_B - \theta_A \leq d \mid \text{data}) = \int_0^1 f_B(t) [1 - F_A(t - d)] dt.$$

Here f_k and F_k are the posterior density and CDF of θ_k .

An exact answer for $P(\theta_B > \theta_A)$. Setting $d = 0$,

$$P(\theta_B > \theta_A \mid \text{data}) = \int_0^1 f_B(t) F_A(t) dt.$$

Both Beta parameters are whole numbers, so f_B and F_A are polynomials in t with rational coefficients. For example, $F_A(t) = \sum_{j=63}^{101} \binom{101}{j} t^j (1-t)^{101-j}$, the binomial-tail form of the Beta CDF. The integral is therefore an exact rational number, which we compute in exact fraction arithmetic:

$$P(\theta_B > \theta_A \mid \text{data}) = 0.882419268177 \dots$$

We obtained the same fraction from an independent finite-sum formula. Numerical quadrature agrees to 10^{-12} . A Monte Carlo run with 2,000,000 posterior draws gives 0.8823, which is -0.71 standard errors from the exact value.

Table 3: Pairwise differences. Bayesian columns are posterior summaries; the last two columns give the classical Wald interval and one-sided z-test of the same difference.

Difference	Post. mean	95% CrI	$P(\delta > 0)$	$P(\delta > 0.05)$	95% Wald CI	One-sided p
$\theta_B - \theta_A$	0.078	(−0.051, 0.207)	0.882	0.667	(−0.051, 0.211)	0.116
$\theta_B - \theta_C$	0.039	(−0.089, 0.167)	0.726	0.435	(−0.089, 0.169)	0.272
$\theta_C - \theta_A$	0.039	(−0.092, 0.170)	0.721	0.437	(−0.093, 0.173)	0.278

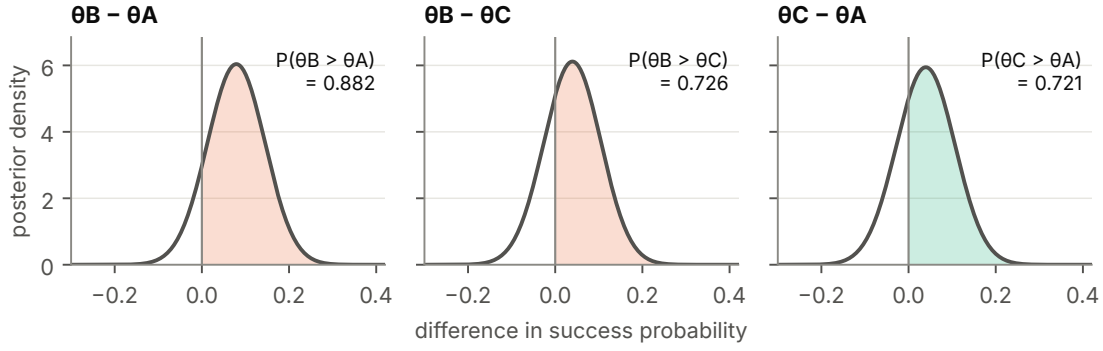


Figure 2: Posterior densities of the pairwise differences. The shaded area to the right of zero is the posterior probability that the first treatment is better.

Reading the differences. Every 95% credible interval for a difference contains zero, so no difference is settled at that level. The Bayesian answer to “is B better than A?” is still direct: probability 0.882. The classical test of the same comparison gives $p = 0.116$, which is “not significant” at 5% and says nothing about how likely B is to be better. Under flat priors and large samples the two numbers are related, since $1 - p = 0.884 \approx 0.882$, but only the posterior probability answers the question the clinician asks. A difference may also be too small to matter. The probability that B beats A by more than 5 percentage points is only 0.667.

5 Which treatment is best? (question 6)

Treatment k is best when θ_k exceeds both others. By independence,

$$P(k \text{ is best} \mid \text{data}) = \int_0^1 f_k(t) \prod_{j \neq k} F_j(t) dt,$$

which is again an exact rational number. We obtain $P(\text{B best}) = 0.680515173383 \dots$, $P(\text{C best}) = 0.248$ and $P(\text{A best}) = 0.071$. The three exact fractions sum to exactly 1, and Monte Carlo agrees with every exact value to within 0.71 standard errors. Figure 3 shows these probabilities.

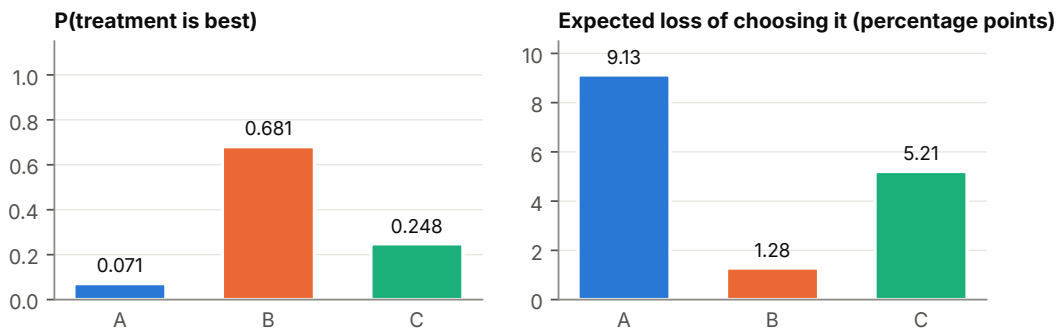


Figure 3: Left: posterior probability that each treatment is the best. Right: expected opportunity loss of choosing each treatment, $E[\max_j \theta_j - \theta_k \mid \text{data}]$, in percentage points.

6 Recommendation (question 7)

Choosing a treatment is a Bayes decision. The Bayes action minimises posterior expected loss [Module 4, Definition 4.3, p. 7]. Two natural losses give the same answer:

- *Opportunity loss* $L(\theta, k) = \max_j \theta_j - \theta_k$ is the success rate given up by not choosing the best treatment. Its posterior expectation is $E[\max_j \theta_j] - E[\theta_k]$, and $E[\max_j \theta_j] = 0.7089$ does not depend on k , so the Bayes action is the largest posterior mean. The expected losses are 1.28 (B), 5.21 (C) and 9.13 (A) percentage points.
- *Zero-one loss* (1 if the chosen treatment is not the best) has posterior expected loss $1 - P(k \text{ best})$, so the Bayes action is the treatment most likely to be best.

Both losses choose B, just as the squared-error and zero-one losses gave the posterior mean and mode for a single θ .

Table 4: Prior sensitivity. The same priors are used for all three treatments.

Prior	$E[\theta_B \text{data}]$	$P(\theta_B > \theta_A)$	$P(\theta_B > \theta_C)$	$P(\text{B best})$	Exp. loss of B	Choice
Uniform Beta(1, 1)	0.696	0.882	0.726	0.681	0.0128	B
Jeffreys Beta($\frac{1}{2}, \frac{1}{2}$)	0.698	0.884	0.728	0.682	0.0128	B
Beta(2, 2)	0.692	0.880	0.724	0.677	0.0129	B
Beta(10, 10)	0.667	0.859	0.707	0.652	0.0135	B

Robustness to the prior. Table 4 repeats the analysis with the Jeffreys prior [Module 4, §5.3.3, p. 14] and the two priors of assignment problems 1.2 and 1.12. Even Beta(10, 10), which is worth 20 imaginary patients centred on 50%, changes $P(\text{B best})$ only from 0.681 to 0.652. The choice is B every time, so the conclusion does not depend on the prior.

How strong is the evidence? B is very probably better than A (0.882). It is only modestly likely to be better than C (0.726), and its 95% credible interval for $\theta_B - \theta_C$ runs from -0.089 to 0.167 . Figure 4 shows how this would change with more patients, under the purely illustrative assumption that the success rates stayed exactly as observed. Then $P(\text{B best})$ would first reach 0.95 at about 750 patients per arm, seven and a half times the present trial. This is not a prediction; real future data will vary.

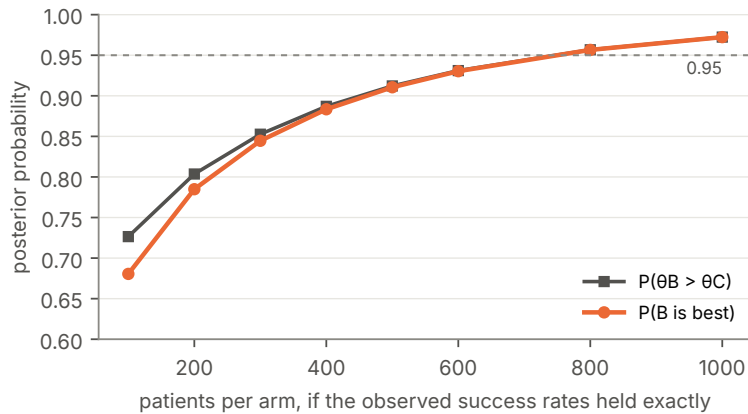


Figure 4: Posterior probabilities for B if each arm had more patients with the same observed success rates (illustration only).

Recommendation. Choose treatment B. It has the highest estimated success rate (0.6961; 95% CrI 0.604–0.781) and the highest probability of being best (0.681). It also carries the smallest expected cost of being wrong: 1.28 percentage points of success rate, against 5.21 for C and 9.13 for A. This holds under all four priors. Because B and C are not clearly separated, a decision with serious consequences, or one where B is more costly or has more side effects, would justify a further trial comparing B with C.

7 Course method check: Lindley and Tierney–Kadane (Module 5)

Module 5 approximates a posterior expectation $E[u(\theta) \mid \mathbf{x}]$ when it has no closed form. Lindley’s method [slides 7–11] expands around the MLE: $E[u] \approx u(\hat{\theta}) + \frac{1}{2}\sigma^2(u_2 + 2u_1\rho_1) + \frac{1}{2}\sigma^4 L_3 u_1$, with $\sigma^2 = -1/L_2(\hat{\theta})$. The Tierney–Kadane method [slides 13–19] uses $E[u] \approx (\sigma_*/\sigma_0) \exp\{n[L_*(\hat{\theta}_*) - L_0(\hat{\theta}_0)]\}$. Our problem has exact conjugate answers, so it measures their error directly: $E[\theta] = a/(a+b)$ and $E[\theta/(1-\theta)] = a/(b-1)$ for a $\text{Beta}(a, b)$ posterior. Before applying the methods, we checked that our implementation reproduces the course’s own worked example: Lindley 1.9965 (slide 12) and Tierney–Kadane 1.995138 (slides 18–19).

Table 5: Module 5 approximations against exact posterior expectations (uniform prior). Error = approximation – exact.

	$E[\theta \mid \text{data}]$			$E[\theta/(1-\theta) \mid \text{data}]$ (odds)		
	Exact	Lindley error	T-K error	Exact	Lindley error	T-K error
A	0.617647	-4.7×10^{-5}	-5.2×10^{-5}	1.657895	0	-6.3×10^{-5}
B	0.696078	-7.8×10^{-5}	-6.2×10^{-5}	2.366667	0	-1.9×10^{-4}
C	0.656863	-6.3×10^{-5}	-5.7×10^{-5}	1.970588	0	-1.1×10^{-4}

Both methods are accurate to about 10^{-4} , the $O(n^{-2})$ order the slides predict for $n = 100$. For the posterior mean, Lindley simplifies to $s/n + (n-2s)/n^2$. For the odds, its three terms add up to

$$\frac{s}{n-s} + \frac{s}{(n-s)^2} + \frac{n-2s}{(n-s)^2} = \frac{s+1}{n-s},$$

which is exactly the conjugate answer, so its error in Table 5 is zero.

8 Limitations

- **Simulated data.** The figures come from the project sheet, not a real trial, and the conclusions apply only to these numbers.
- **Independence and a common rate.** The model assumes that patients within an arm are independent with a common success probability, and that the arms are independent. Different patient populations across arms, clustering (for example by hospital) or a non-randomised allocation would break this.
- **One outcome.** Success is the only outcome. Costs, side effects and severity of failure are ignored, and any of them could change the decision.
- **No pooling across treatments.** Independent priors treat the three rates as unrelated. A hierarchical model would let them share information, but it is a multi-parameter model outside the single-parameter course syllabus.

9 Reproducibility and verification

Every computed result in this report is written by `scripts/run_analysis.py` into a file that the report reads, so no result was copied by hand. The remaining figures in the text (the data, the course’s worked-example values and descriptive ranges such as “ ± 9 percentage points”) are checked separately in `checks/test_report.py`. Each exact result was verified by a second exact method (the finite-sum formula, the binomial-tail form of the CDF, or a second integration order), by quadrature and by seeded Monte Carlo judged by its standard error. The conjugate posterior was also checked against Bayes’ theorem evaluated numerically on a grid. The checks run with `python checks/run_checks.py`.

A Course mapping

Topic used	Course source
Bernoulli and binomial distributions	Module 1, §1.2–1.3, pp. 5–9
Normal distribution, Φ (Wald intervals)	Module 1, §2.5, pp. 24–29; assignment 1.7–1.8
Independence of the three arms	Module 2, Definition 2.8, p. 9
Sufficiency of s (Neyman–Fisher factorisation)	Module 3, §2, p. 4
MLE $\hat{\theta} = s/n$	Module 3, p. 14
Beta–binomial conjugate posterior	Module 4, Examples 4.6, 4.9; §5.1.1, pp. 4–11
Posterior mean, median, mode as Bayes estimators	Module 4, Theorems 4.1–4.3, pp. 7–8
Bayes action minimising posterior expected loss	Module 4, Definition 4.3, p. 7
Jeffreys prior $\text{Beta}(\frac{1}{2}, \frac{1}{2})$	Module 4, §5.3.3, p. 14
Lindley and Tierney–Kadane approximations	Module 5, slides 7–19
Simulation from the posterior, $P(\theta_B > \theta_A)$	Module 5, slide 20; assignment 1.10
Credible intervals, prior sensitivity, predictive probability	Assignment problems 1.2, 1.11, 1.12